

Expt 3. Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

```
1  clear; close all;
2
3  % Specifications
4  Fs=2000; Rp=1; Rs=40;
5  Wp=1000/(Fs/2);
6  Ws=2000/(Fs/2);
7
8  % Butterworth Filter
9  [b1,a1]=butter(Wp,Ws,Rp,Rs);
10 [b2,a2]=butter(b1,a1,'bandpass');
11
12 % Chebyshev Type-I Filter
13 [b2,a2]=cheb1ord(Wp,Ws,Rp,Rs);
14 [b2,a2]=cheby1(b2,a2,'bandpass');
15
16 % Magnitude Response
17 figure;
18 freqz(b1,a1); title('Butterworth Filter');
19
20 figure;
21 freqz(b2,a2); title('Chebyshev Type-I Filter');
22
23 % Group Delay
24 figure;
25 grpdelay(b1,a1); hold on;
26 grpdelay(b2,a2);
27 legend('Butterworth','Chebyshev-I');
28
29 % Pole-zero Plot
30 figure;
31 subplot(121),zplane(b1,a1); title('Butterworth');
32 subplot(122),zplane(b2,a2); title('Chebyshev-I');
33
34 % Impulse Response
35 figure;
36 subplot(211),impz(b1,a1); title('Butterworth Impulse');
37 subplot(212),impz(b2,a2); title('Chebyshev Impulse');
38
39 % Pole Magnitude
40 figure;
41 subplot(121),stem(abs(roots(a1)),'filled');
42 title('Butterworth Pole Magnitude');
43
44 subplot(122),stem(abs(roots(a2)),'filled');
45 title('Chebyshev Pole Magnitude');
```

Command Window

New to MATLAB? See resources for [Getting Started](#)

Butterworth Order = 9

Chebyshev Order = 5

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